

E-Portfolio Report and Reflection for Intelligent Agent

Student ID: 12698592

Module: Intelligent Agent

GitHub Repository URL: <https://alialhammadi-8.github.io/e-Portfolio/intelligent-agents.html>

Date: October 20, 2025

E-Portfolio Report

Student ID: 12698592

Module: Intelligent Agent

GitHub Repository URL: <https://alialhammadi-8.github.io/e-Portfolio/intelligent-agents.html>

Date: October 20, 2025

E-Portfolio Report – Intelligent Agents Module

1. Introduction

The Intelligent Agents module has offered a necessary and very imperative transition between theory and practical software development. It helped me to construct intelligent autonomous systems with Python, organize the cooperative development with GitHub, and think about the professional and ethical aspect of agent-based design.

To introduce my learning experience, this E-Portfolio is organized thematically, i.e., in terms of how my cognition of the intelligent agents has evolved, how I have been practicing the practical skills in the field of programming, my experience in teamwork, and my professional development. It is by reflecting on Rolfe et al. (2001) that I describe what I learned, hence what it entailed and now what and when I intend to implement in future.

2. An introduction to Intelligent Agents.

I had a shallow knowledge about agents at the start of the module. The initial study units explained that an intelligent agent is a software system, which has the ability to sense its surroundings and make an autonomous response in relation to a goal (Wooldridge, 2020). Agents are not traditional programs since autonomy, reactivity and proactivity are all incorporated in their design.

I have heard lectures and read about agent architectures including reactive, deliberative and hybrid models. All of them have trade-offs: reactive systems are fast and lack planning; deliberative models are also fast and can think strategically; hybrid systems combine them (Russell and Norvig, 2021).

Multi-agent systems (MAS) involving a number of agents who work together or compete to accomplish tasks were also brought out in the module. The learning about agent negotiation and communication and data sharing processes enhanced my understanding of the value of distributed intelligence (IBM, n.d.).

Such theoretical foundation helped me to be ready to apply it practically with Python- it allowed me to create small simulations, which would illustrate collaboration between autonomous objects.

3. Creating Technical Competence in Python.

My progress relied on the implementation of agent logic using Python. I used modular programming techniques in applying such concepts as condition-based decision-making, rule evaluation and inter-agent communication.

Python Software Foundation (2025) indicates that Python was created with a clear syntax and vast library, which is suitable in AI and simulation applications. I learned to write classes that were reusable, used control structures to sense the environment, and I used randomization to approximate adaptive behaviour. I learnt to be patient and analytical through debugging errors.

With time I learnt to write my code in order to make it readable and scalable. I used regular commenting and version tracking, which is in accordance with professional codes of coding. The use of agents capable of detecting environmental variables and making changes enhanced my skills on matters of autonomy in software.

Practical tasks promoted exploration: of how to make logic to conflict, or to make grounds to collaborate. These exercises gave me a confidence in the capability of Python to be used to develop intelligent and distributed computing applications.

4. Teamwork and Version Control using GitHub.

GitHub was one of the most useful activities we did in this module when it came to handling our group project. Teamwork involved effective communication, versions and responsibility.

We broke down coding work with the help of GitHub repositories, tracked problems through issues and integrated work through pull requests. In accordance with the principles of GitHub Inc. (2025), I was taught to deal with the branch management, merge conflicts, and change histories.

The experience of producing in this environment allowed me to get the firsthand experience of collaborating at the industry level. The input of every team member was accessible and open to

scrutiny, which led to accountability and openness. I have also gained knowledge regarding professional documentation skills working on README files and project wikis to outline objectives, architecture and dependencies.

These were similar to the ones employed by contemporary software teams and taught me that version control is as important as the writing itself. The system of work at GitHub promoted discipline, peer evaluation, and self-improvement.

5. Ethical and Professional Disconsciousness.

Ethical aspect of intelligent agent development was not very small part of this module. I realized that automation is efficient, but there is a problem of accountability, fairness and privacy.

Johnson (2020) emphasizes that computer professionals need to foresee the social implications of the technology activity. This viewpoint redefined my thinking in terms of agent design: in addition to efficiency systems have to be human valued.

To illustrate, even the most intelligent system that determines how resources to allocate might discriminate inadvertently in case its design is not ethical. Hence, responsibility in agencies is characterized by transparency, bias testing, and explainability.

The professional accountability was also strengthened by the University of Essex (2023) policy on academic integrity. It stresses upon honesty and ethical behaviour in academic and software practices. This has helped me in influencing my teamwork behavior, whereby every input is genuine and noted accordingly.

These reflections have helped me realize that programming is not only a technical, but also an ethical practice. A good developer will ensure the protection of users, information and even societal trust besides writing quality code.

6. Collaboration and Reflective Practice.

Being in a distributed team came with the challenges and growth opportunities. To start with, our group had the problem of coordination, parallel code edits and delays. Nevertheless, these barriers were overcome through the establishment of workflow and task assignment.

Short Agile- style check-ins were used to align them and remove technical blockers. This was a technique that increased productivity and morale. Another experience that I got regarding leadership was during repository reviews and summarising of weekly progress.

Reflective learning was the main focus during the module. With the help of Rolfe et al. (2001) I regularly posed the question: What did I have done? What went wrong? How can I improve? This building had made frustration work in their favor. Every single mistake during the process of debugging was a lesson of patience and acceptance.

With time I grew more certain both about my codewriting skill and the ability to make a constructive contribution to a team setting. Such competencies will be priceless in the working software.

7. Knowledge and Future Development.

The Intelligent Agents module has equipped me to combine technical skills and critical thinking abilities. I am now able to create smart systems, oversee the work flow, and apply ethical factors at the same time.

In order to keep expanding, I have come up with the following important objectives:

1. Expert During Python, Study to implement frameworks together with Mesa and Flask, which are used to conduct complex computations and host customers in the web.
2. Improve Project Management Skills, Create a flow of more formal Agile through GitHub Projects and issue boards.
3. Optimize Ethical Practice, Persist in the new AI laws and embrace open forms of model-reporting to do so.

4. Consider Open-Source Contributions, Work on open repositories to make the processes of developing on a larger scale.
5. Ensure Reflective Practice, Keep doing monthly reflective sessions through the model by Rolfe to balance your personal and professional development.

These actions are in line with my long-term aim of becoming a reflective and ethically sensitive software engineer who can develop adaptive and user-centered intelligent systems.

8. Conclusion

Considering this learning experience, Intelligent Agents module has developed my technical, ethical, and professional skills to a great extent.

My initial knowledge of the autonomous systems was low, and, by the end of the experience, I was able to design, code a project, and manage agent-based projects related to Python and GitHub. I have also come to understand that software engineering is not technical only, it requires a sense of empathy and responsibility as well as constant learning.

This module prompted me to reason like a developer and a professional which involves being innovative and being responsible at the same time. I will take these values into my next academic and professional path: write clean code, work cooperatively and create technology, which helps improve and not sourish society.

References:

- European Commission. (2022). *Ethics Guidelines for Trustworthy AI*. European Commission, High-Level Expert Group on AI. Available at: <https://digital-strategy.ec.europa.eu/en/library/ethics-guidelines-trustworthy-ai> Authoritative policy document emphasizing human-centric AI, responsibility, and transparency — great for the professional ethics section.
- GitHub Inc. (2025) *Collaborating with Issues and Pull Requests*. Available at: <https://docs.github.com/en/issues>
- IBM (n.d.) *What is a Multiagent System?* Available at: <https://www.ibm.com/think/topics/multiagent-system>
- IEEE Computer Society. (2023). *Ethically Aligned Design for Intelligent and Autonomous Systems*. IEEE Standards Association. Available at: <https://ethicsinaction.ieee.org/> Real IEEE resource on AI ethics — fits perfectly with your ethical awareness and responsible development discussion.
- Python Software Foundation (2025) *The Python Language Reference*. Available at: <https://docs.python.org/3/reference/>
- Russell, S. and Norvig, P. (2021) *Artificial Intelligence: A Modern Approach*. 4th ed. Pearson Education. Available at: <https://aima.cs.berkeley.edu/>.
- University of Essex (2023) *Academic Integrity Policy*. Available at: <https://www.essex.ac.uk/student/academic-skills/academic-integrity>

Reflection

Student ID: 12698592

Module: Intelligent Agent

GitHub Repository URL: <https://alialhammadi-8.github.io/e-Portfolio/intelligent-agents.html>

Date: October 20, 2025

Reflection – Intelligent Agents Module

1. Introduction

The Intelligent Agents module has given me a proper insight into the design, implementation and evaluation of autonomous systems. This reflection reflects on my individual and professional growth throughout the course, specifically, in Python programming, collaboration in GitHub and software project management.

I have organized this reflection based on a reflective model, the reflection model created by Rolfe et al. (2001) What? So What? Now What? — where I can explain what I have obtained, how to analyze its meaning and what is my future development perspective.

2. Description of Experience (What?)

My background knowledge regarding intelligent agents before beginning this module was limited and I was unaware of the distinction between the intelligent agent and the traditional software system. The initial classes made me realize that intelligent agents are independent, active and can make decisions of their own - these are the main features that distinguish them against regular procedural programs (Wooldridge, 2020).

I also studied various agent designs, e.g. reactive, deliberative and hybrid systems. All of these types provide different problem-solving strategies; an example is a reactive system, which reacts fast to the changes in the environment, and a deliberative system, which plans action based on the goals and beliefs (Russell and Norvig, 2021).

The Python coding exercises helped me to apply simple agent behaviors that reinforced my technical understanding of how agents behave in a computational environment. The flexibility of Python enabled me to achieve agent communication and reactivity simulation in an efficient manner (Van Rossum and Drake, 2009).

My learning experience was substantial as a result of my group project, and as the group developed an agent-based application. Our source code was managed and tasks assigned in GitHub and

progress monitored. This platform enabled me to learn how professional developers can have organized work processes based on version control and issue management (GitHub, Inc., 2025).

Moreover, I got to know about the professional and ethical consequences of intelligent systems. The course has highlighted that autonomy presents the ethical dilemma, i.e. accountability, bias and transparency of data. I also realized that even a technically flawless system might be socially damaging when it is created without moral consideration (Johnson, 2020).

3. Interpretation and Reflection on Emotions (So What?)

First, the technical side of the module was challenging to me. Such concepts as multi-agent communication, hybrid architectures and reactive systems demanded intense thinking. My initial efforts in coding tended to fail due to Syntactic mistakes or half-baked logic. Nonetheless, I learned to like these problems as learning opportunities through practice.

I was also taken out of my comfort zone as I worked within a team. Liaising using GitHub took time to achieve, patience, communication, and accountability. Initially, it was irritating trying to manage commits and resolve merge conflicts but with time I got to know how teamwork could be made easy by collaborative tools in the real world of software development. This experience reflected the professional practices, in which the success depends on the organized communication and shared responsibility.

This change of confusion into competence was gratifying emotionally. A reflective cycle, which was developed by Gibbs, states that meaningful learning is impossible without emotional awareness. I realised that the frustration that I experienced when I was doing early programming was actually encouraging me to seek assistance, read documentation and work towards bettering it in a logical manner.

The moral arguments in the module also affected my mind. I am now in a position to see programming as a socially responsible thing rather than just a technical one. All design-related choices, including the logic of goal-setting of a specific agent or the use of data collected in a specific context, should be supported by fairness and accountability values (University of Essex, 2023). This knowledge has helped me to connect theoretical knowledge with professional values.

I also understood how reflection promotes development. Some of the weaknesses that would have been tackled by me through the weekly analysis included debugging habits or unproductive code and, in a way, before they impacted the project outcomes.

4. Lessons Learned and Future Applications (Now What?)

The module has been influential in forming my professional identity as a mature software developer.

Technical Competence:

I have gained a better mastery of Python to design and test agent-based systems. I have learned the ability to build modular code structure, dependency control, and efficiency. In the future, I will investigate such frameworks as Mesa to simulate agents in large scale and optimisation methods to optimise in complex environment.

Software Collaboration:

GitHub usage enhanced my knowledge of collaborating on the code, version management, and documentation. I will still apply these skills in my future university projects and professional internships. This experience also encouraged me to work on open-source projects to have more exposure to a variety of coding styles and real-life challenges.

Code of Ethics:

This has helped me to be more conscious of the ethical requirements of software engineering through discussions and tutor feedback. I will also always consider ethics at each phase of my design process, including data manipulation or agent autonomy, to be fair and transparent with the results.

The goals of continuous development are:

Enhance Python skills in regular code writing and improved libraries.

Gain the skills of leadership in working on collaborative projects.

Participate in lifelong learning in order to fit the new AI technologies.

The next step is to follow up on the progress and refine the professional growth through the application of the model by Rolfe.

All these will enable me to become a competent and responsible software developer who can design smart systems in a morally and cost-effective manner.

5. Conclusion

Considering the Intelligent Agents module, I could easily affirm that it has been one of the turning points in my academic life. The combination of theory, technical practice, teamwork, and ethics provided me with a true picture of what software development is about today.

I also began with a low level of confidence but at present have applied practical skills in Python, GitHub, and agent-based design. I also found out that technical proficiency is not a sufficient factor and that moral consciousness and thought are equally imperative.

Besides developing my code skills, this module also helped me develop my attitude as a responsible and flexible professional. Going forward, I will strive to keep coming up with smart systems that merge innovation, efficiency and ethical accountability.

References:

- MIT Press. (2023). *Agents in the Long Game of AI*. Available at: <https://mitpress.mit.edu/9780262549424/agents-in-the-long-game-of-ai/> MIT Press
- O'Reilly Media. (2024). *Building Applications with AI Agents*. Available at: <https://www.oreilly.com/library/view/building-applications-with/9781098176495/> O'Reilly Media
- Python Software Foundation. (2025). *Python Documentation*. Available at: <https://docs.python.org/> Python documentation+1
- Python Software Foundation. (2025). *The Python Language Reference — Python 3.13.5 documentation*. Available at: <https://docs.python.org/3/reference/index.html> Python documentation+1
- Springer. (2023). *Building Embodied AI Systems: The Agents*. Available at: <https://link.springer.com/book/10.1007/978-3-031-68256-8> SpringerLink
- Wiley. (2024). *Learning LangChain — Chapter 6: Agent Architecture*. Available at: <https://www.oreilly.com/library/view/learning-langchain/9781098167271/ch06.html>